ASEAN
GEOAI FUSION 2026

Module 7: Advanced Knowledge: Advanced Geospatial Techniques in GeoAI Applications (AD1002)

AGAIF 2026 Boot Camp 2
Level: Advanced Knowledge

23 July 2026



Session 2:

Time Series Analysis for Networks

Session Agenda

Overview of the key topics we will explore in this session

01 What is Time Series Analysis for Network

02 Temporal patterns in network data

03 Trend Analysis and seasonability

04 Forecasting techniques

05 Deep Learning methods for time-series prediction

Learning Outcomes



1. Explain time-series analysis and temporal patterns in network data.



2. Differentiate between trend and seasonality in network measurements.



3. Compare common forecasting techniques for network time-series data.



4. Explain how LSTM uses previous multivariate observations to predict the next RSRP.



5. Connect predicted RSRP with geographic coordinates for GeoAI-based signal mapping.

Opening Notes

- Networks continuously produce data such as traffic, bitrate, latency and signal strength.
- When these measurements are recorded in time order, they form network time-series data.
- In this session, we will follow one complete process:
  - starting with network measurements,
  - identifying temporal patterns,
  - analysing trends,
  - selecting a forecasting method and finally using LSTM to predict future network performance.
- We will use a real 4G/5G dataset collected around Sunway University, Selangor as an example to visualize the concept.

Case Study: Predicting Mobile Signal Strength

Real 4G/5G network measurements collected around Sunway University, Selangor

What is in the dataset?

- 30,925 observations
- 23 measurement sessions
- Walking and driving
- Operators A, B and C

One row = network condition at one time and location

What are the input features?

- Previous Level (RSRP)
- SNR
- CQI
- DL_bitrate
- UL_bitrate
- Speed
- Latitude
- Longitude

What does the model predict?

10 previous observations
8 input features

→ LSTM →

Next Level (RSRP)
Prediction target



Location-aware GeoAI map shows where strong or weak signals may occur.

Important: The dataset supports short-term, location-aware signal prediction along a route, not daily seasonality analysis.

How the Case Study Connects to This Session

We begin with the network problem, then build the knowledge needed to solve it.

1

Observe the network

RSRP and other measurements are recorded in time order during each drive-test session.

2

Recognise patterns

We examine how measurements increase, decrease or change suddenly over a short sequence.

3

Predict the next value

The model uses previous observations to estimate the RSRP at the next observation.

4

Map the prediction

Latitude and longitude connect each prediction to its geographic location.

What Is Time-Series Analysis for Networks?

Time-series analysis for networks examines network measurements recorded in time order to identify patterns and predict future network conditions.

Examples of network time-series data

- Network traffic
- Download and upload bitrate
- Latency and packet loss
- Bandwidth utilisation
- Mobile-signal strength

Past network data → Identify patterns → Predict future conditions

Why Time-Series Analysis for Networks?



1. Understand how network behaviour changes over time and predict future conditions.



2. Prevent congestion, optimise resources, reduce costs, and improve service quality.



3. Identify temporal patterns such as trends, seasonality, and recurring behaviour.



4. Forecast network traffic and support better resource planning and network performance.

Temporal Patterns in Network Data

Temporal patterns show how network measurements change or repeat over time.

Common patterns

- Increase
- Decrease
- Stable
- Sudden change
- Peak and fall

Timestamped measurements

↓

Identify patterns

↓

Understand network behaviour

Download bitrate example



The bitrate increases, reaches a peak, and then decreases.

Malaysian dataset: DL_bitrate and UL_bitrate change during HTTP, FTP and video-streaming sessions.

Trend vs Seasonality in Network Data

Does the network measurement move in one direction, or does the same pattern repeat?

TREND

The overall direction of network data during a period.
It does not need to repeat.

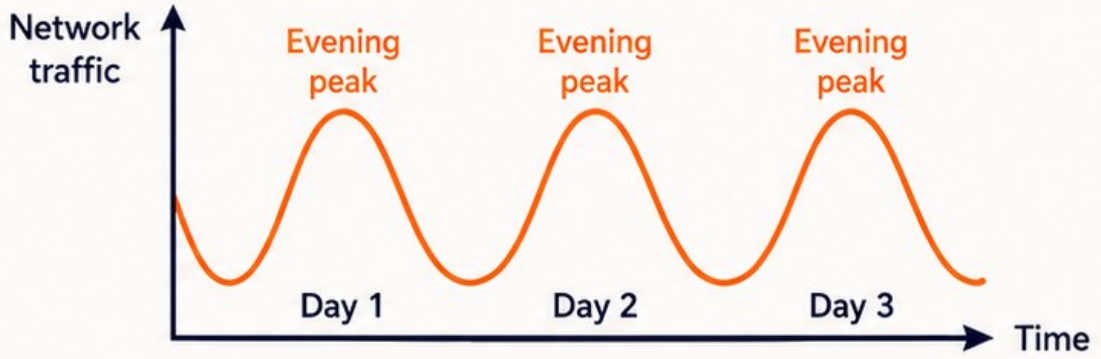


Example: Signal strength gradually decreases as a user moves along a route.

Direction changes over time

SEASONALITY

The same network pattern repeats after a fixed interval, such as every hour or every day.



Example: Network traffic reaches a similar peak every evening.

Pattern repeats regularly

What can this dataset analyse?

✓ **Short-term trend:** Yes — measurements are recorded in time order within each journey.

✗ **Seasonality:** No — the dataset contains separate short journeys, not a continuous time series.

Seasonality requires the same network measurement to be collected repeatedly at fixed intervals across many cycles, such as hourly patterns repeated over several days.

Forecasting Techniques for Time Series Data

Category	Techniques	Suitable for
Statistical Methods	Naïve, Moving Average, Exponential Smoothing, ARIMA	Simple or linear time-series
Machine Learning Methods	Linear Regression, Random Forest, XGBoost	Nonlinear relationships with feature engineering
Deep Learning Methods	LSTM, GRU, Transformer	Complex sequential and multivariate data

There is no single forecasting technique that works best for every dataset.
The choice **depends on the characteristics of the data.**

Our network dataset contains multiple variables collected continuously over time.
Which category is the **most suitable?**

Why Move from Traditional Forecasting to Deep Learning?

The forecasting method should match the complexity of the network data.

1

Multiple variables

RSRP is influenced by related measurements such as RSRQ, RSSI, CQI, bitrate and movement.

2

Nonlinear changes

Signal strength may change irregularly because of distance, obstruction, handover and mobility.

3

Ordered sequences

The meaning of the current observation depends partly on measurements recorded before it.

Deep learning is considered because it can learn complex relationships directly from multivariate sequences.

What Is Deep Learning?

Understanding the foundation of modern time-series forecasting

AI HIERARCHY

- **Artificial Intelligence**
- ↓
- **Machine Learning**
- ↓
- **Deep Learning**

1. DEFINITION

Deep learning is a branch of machine learning that uses artificial neural networks with multiple hidden layers to **automatically learn complex patterns from data.**

2. WHY DEEP LEARNING?

Capability	Simple Idea	Network Example
Learns complex relationships	Learns nonlinear relationships among multiple variables.	Models the relationship between RSRP, SNR, CQI, throughput, and speed.
Learns from sequential data	Uses previous observations to understand temporal behaviour.	Uses historical network measurements to predict future network conditions.
Automatically learns features	Identifies important patterns without extensive manual feature engineering.	Reduces the need to manually select influential network features.
Supports forecasting	Estimates future values based on learned patterns.	Predicts future RSRP or network traffic.

3. WHY IS DEEP LEARNING SUITABLE FOR OUR DATASET?

Temporal
Timestamp

Spatial
Latitude, Longitude

Network Performance
RSRP, SNR, CQI, DL/UL Bitrate

Mobility
Speed

KEY MESSAGE

Our dataset is **sequential, multivariate,** and **spatio-temporal.** Deep learning is well suited to **learning complex temporal patterns** from this type of data.

Deep Learning Models for Time Series Forecasting

Model	Strength	Limitation	Suitable For
 **RNN**	Basic sequential learning	Short memory	Short sequences
 **LSTM**	Long-term memory	Higher computational cost	Long sequential data
 **GRU**	Faster than LSTM	Simpler memory mechanism	Medium-length sequences
 **Transformer**	Parallel processing	Requires large datasets	Very long sequences



Each model is designed to process sequential data differently.
The most suitable model depends on the **characteristics of the problem.**

Understanding Our Network Dataset

1. FEATURE GROUPS

Feature Group	Variables	Purpose
Temporal	Timestamp	Records when measurements are collected
Spatial	Latitude, Longitude	Identifies where measurements are collected
Network Performance	RSRP, SNR, CQI, DL Bitrate, UL Bitrate	Represents network quality and performance
Mobility	Speed	Captures user movement that affects network conditions

2. DATASET CHARACTERISTICS

Characteristic	Explanation
Sequential observations	Measurements are recorded continuously over time.
Multiple network variables	The dataset contains several network measurements collected simultaneously.
Nonlinear behaviour	Network conditions change dynamically due to user movement and environmental factors.
Continuous measurements	Data is collected continuously, allowing temporal pattern analysis and forecasting.

Key Message: Our dataset is a spatio-temporal, multivariate time-series dataset containing **temporal**, **spatial**, **network performance**, and **mobility** information.

Why Is LSTM Suitable for Our Dataset?

The dataset characteristics match the main strengths of an LSTM model.

1

Time ordered

Every measurement has a timestamp and belongs to a measurement session.

2

Multivariate

The model can learn from several network variables recorded at the same observation.

3

Sequential context

Ten previous observations provide context for predicting the next RSRP value.

4

GeoAI output

Predictions can be joined to coordinates and displayed on a signal map.

Model task: use the previous 10 observations to predict the RSRP at the next observation.

What is LSTM?

DEFINITION

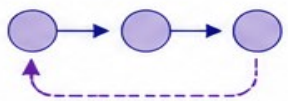
Long Short-Term Memory (LSTM) is a type of **Recurrent Neural Network (RNN)** designed to learn **long-term dependencies** in sequential data.

It remembers important historical information while forgetting irrelevant information, making it highly effective for **time-series forecasting**.

RNN vs LSTM

RNN

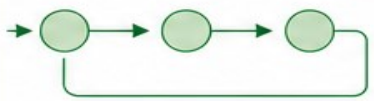
Remembers short-term information only.



Struggles with long sequences

LSTM

Remembers long-term information effectively.



Captures long-term dependencies

LSTM in a Nutshell

LSTM uses past observations to predict the next future value.

Historical Network Measurements

RSRP | SNR | CQI | DL Bitrate | UL Bitrate | Speed

t-4 → t-3 → t-2 → t-1 • • •

LSTM (Model)

Future Prediction

Predict Next RSRP (at time t)

KEY FEATURES

Learns from sequential observations

Understands patterns based on the order of time.

Remembers long-term information

Keeps important information for a long time.

Handles nonlinear relationships

Captures complex and nonlinear patterns in the data.

Processes multiple input variables

Learns relationships among multiple network metrics.

KEY MESSAGE

LSTM learns from **historical observations** to predict **future network conditions**.

How Does LSTM Work?

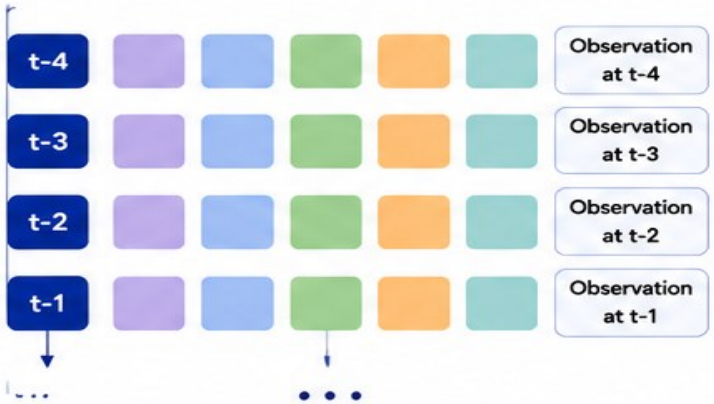
LSTM (Long Short-Term Memory) processes a sequence of data using **four main components** to capture long-term dependencies and generate accurate predictions.

1. Receive a Sequence of Historical Observations

LSTM receives a sequence of past network measurements at different time steps.

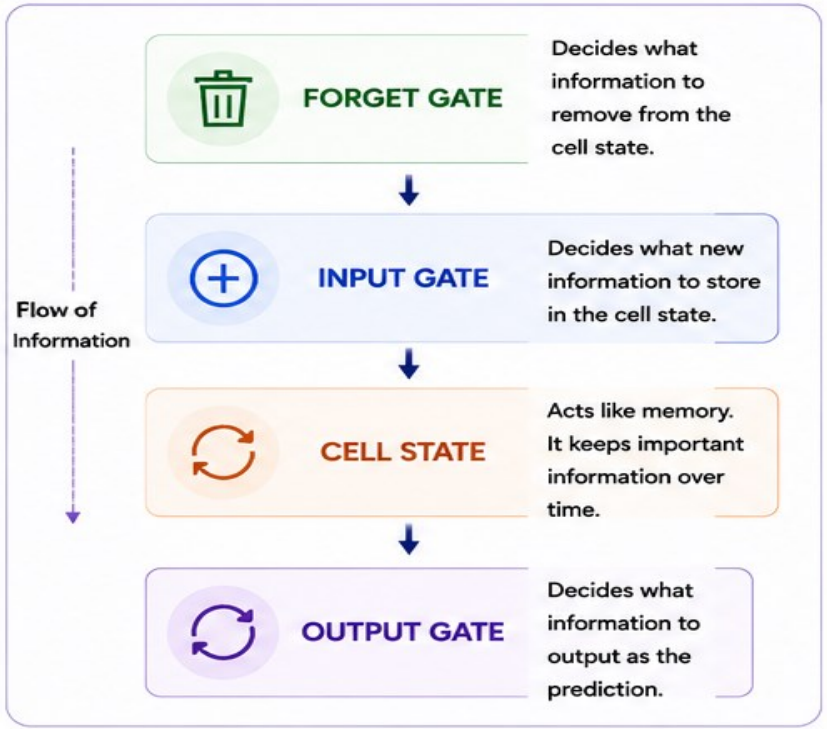
Example Input Features

RSRP (dBm)	SNR (dB)	CQI (Index)	DL/UL Bitrate (Mbps)	Speed (km/h)



2. LSTM Processes the Sequence Using Four Components

Each component plays a specific role to decide what to keep, what to update, and what to output.



3. Generate the Prediction

After processing the sequence, LSTM outputs the prediction for the next time step.

Prediction Output

Predict Next RSRP (at time t)



-99.4 dBm
(Example)

In summary:

LSTM takes past observations → selectively remembers important information → updates its memory → and produces the best prediction for the next time step.

Forget Gate	:	Removes irrelevant information
Input Gate	:	Stores new important information
Cell State	:	Maintains long-term memory
Output Gate	:	Produces the final prediction

LSTM Prediction Workflow with Our Dataset

Goal: Use historical network measurements to predict the next RSRP value for the next time step.

1 Load Dataset

Import the dataset containing network measurements over time.

Dataset Columns:
- Timestamp
- RSRP (target)
- SNR
- CQI
- DL/UL Bitrate
- Speed

2 Preprocess Data

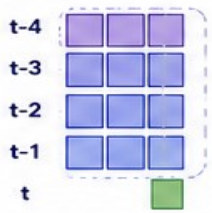
- Handle missing values
- Sort by timestamp
- Scale/normalize features
- Create sequences using past observations

Scaling:
Min-Max / Standard

3 Create Sequences

Use a sliding window of past observations to form input sequences.

Window Size = N



Input: [t-N, ..., t-1]
(Target: RSRP at time t)

4 Train LSTM Model

- Build LSTM architecture
- Train on historical sequences
- Optimize parameters

Loss Function:
MSE (Mean Squared Error)
Optimizer: Adam

5 Evaluate Model

Evaluate model performance using test data.

Evaluation Metrics:
- RMSE
- MAE
- MAPE
- R^2 Score

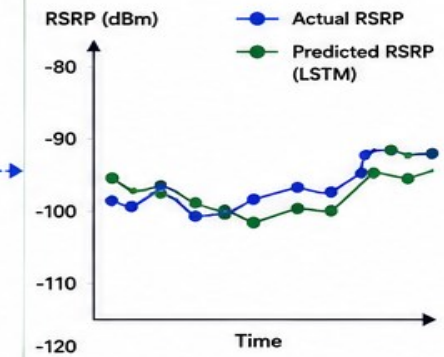
6 Predict Next RSRP (at time t)

Use the latest N observations to predict the next RSRP value.

Input: [t-N, ..., t-1]
Output: RSRP at time t

Example Prediction Output

True vs Predicted RSRP



Next Prediction (at time t)

-99.4 dBm

Interpretation

The model predicts the RSRP value at the next time step based on learned patterns from historical data.

Timestamp	RSRP (dBm)	SNR (dB)	CQI (Index)	DL Bitrate (Mbps)	UL Bitrate (Mbps)	Speed (km/h)
10:00:00	-99.4	18.5	9	45.2	12.1	32
10:00:01	-100.2	17.8	9	40.7	11.4	29
10:00:02	-98.7	19.1	10	46.3	13.0	34
...	...	...	...	...	...	...

End-to-End Workflow

Raw Data → Preprocessing & Scaling → Sequence Creation → LSTM Training → Model Evaluation → Prediction → Better Network Planning

Key Message:
LSTM learns temporal patterns from historical network measurements and accurately predicts the next RSRP value, helping network operators plan capacity and improve user experience proactively.

Benefits
- Early congestion detection
- Optimized network planning
- Improved user experience

Session Review: From Network Data to a GeoAI Prediction

The complete process can be remembered in four connected stages.

1

Prepare

Clean the data, keep each session in time order and scale the selected network features.

2

Create sequences

Group every 10 previous observations as one LSTM input sequence.

3

Train and evaluate

Predict the next RSRP and assess performance using MAE, RMSE and R^2.

4

Map the result

Attach each prediction to latitude and longitude and display it using Folium.

Key idea: time-series prediction becomes GeoAI when the predicted value is connected to geographic location.

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THANK YOU

Bootcamp and Hackathon for ASEAN Geospatial Innovation 2026